

Breast Cancer Prediction in Machine Learning Using Logistic Regression and KNN Algorithms

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Abstract— Breast cancer is one of the leading causes of death among women worldwide. Early detection and accurate diagnosis are essential for improving survival rates and reducing treatment costs. Machine learning techniques have become highly effective in medical diagnosis due to their ability to analyze large datasets and identify hidden patterns. This paper presents a comparative analysis of two supervised machine learning algorithms, Logistic Regression (LR) and K-Nearest Neighbors (KNN), for breast cancer prediction. The study involves data preprocessing, feature scaling, feature selection, hyperparameter tuning using GridSearchCV, and evaluation using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC curve. Experimental results demonstrate that both models achieve excellent predictive performance, with accuracy levels reaching approximately 98–99%. The findings indicate that machine learning can significantly assist healthcare professionals in early breast cancer detection.

Keywords— Breast Cancer Prediction, Machine Learning, Logistic Regression, K-Nearest Neighbors, Feature Selection, Classification.

I. INTRODUCTION

Breast cancer is one of the most dangerous and frequently diagnosed cancers among women globally. According to medical research, early diagnosis can greatly improve patient survival rates and treatment effectiveness. Traditional diagnostic methods often require expert analysis and can sometimes result in delayed or inaccurate predictions. Therefore, the use of intelligent computational techniques has become increasingly important in healthcare systems.

Machine learning (ML) algorithms provide automated methods for identifying patterns in medical datasets and making accurate predictions. These techniques help reduce human error, improve diagnosis speed, and support clinical decision-making. In recent years, supervised learning algorithms have shown promising results in cancer classification tasks.

This study focuses on evaluating the effectiveness of Logistic Regression and K-Nearest Neighbors algorithms for breast cancer prediction. The proposed system applies preprocessing, scaling, feature selection, and hyperparameter optimization to improve classification performance.

II. LITERATURE REVIEW

Several researchers have applied machine learning techniques for cancer prediction and diagnosis. Logistic Regression is widely used due to its simplicity, interpretability, and strong classification performance. KNN is another popular algorithm that classifies samples based on similarity measures and neighboring data points.

IV. MACHINE LEARNING MODELS

A. Logistic Regression

Logistic Regression is a supervised classification algorithm commonly used for binary classification problems.

```
from sklearn.linear_model import
LogisticRegression
from sklearn.model_selection import
GridSearchCV
lr = LogisticRegression(max_iter=10000)
param_lr = {
    'C': [0.001, 0.01, 0.1, 1, 10],
    'penalty': ['l1', 'l2'],
    'solver': ['liblinear']}
}
grid_lr = GridSearchCV(lr, param_lr, cv=10)
grid_lr.fit(X_train, y_train)
```

Advantages: Simple and interpretable, fast training process, and effective for binary classification.

B. K-Nearest Neighbors (KNN)

KNN is a distance-based classification algorithm that predicts classes based on neighboring samples.

```
from sklearn.neighbors import
KNeighborsClassifier
knn = KNeighborsClassifier()
param_knn = {
    'n_neighbors': list(range(1, 21)),
    'weights': ['uniform', 'distance'],
    'metric': ['euclidean', 'manhattan']}
}
grid_knn = GridSearchCV(knn, param_knn,
cv=10)
grid_knn.fit(X_train, y_train)
```

Previous studies demonstrated that preprocessing techniques such as normalization, feature scaling, and feature selection significantly improve prediction accuracy. GridSearchCV and cross-validation methods are commonly used for hyperparameter optimization to achieve optimal model performance.

Research findings indicate that combining feature selection with classification algorithms can increase prediction efficiency and reduce computational complexity. Most studies achieved classification accuracy above 95%, proving the reliability of ML models in medical applications.

III. PROPOSED METHODOLOGY

The proposed methodology includes the following stages:

- Dataset collection and preprocessing
- Feature and target separation
- Feature scaling using StandardScaler
- Feature selection using SelectKBest
- Model training using Logistic Regression and KNN
- Hyperparameter tuning using GridSearchCV
- Performance evaluation using classification metrics

A. Dataset Collection

The breast cancer dataset contains various tumor-related features and diagnosis labels. The diagnosis column consists of M → Malignant and B → Benign. The dataset is loaded using the Pandas library.

```
import pandas as pd
data =
pd.read_csv("breast_cancer_dataset.csv")
```

B. Data Preprocessing

Data preprocessing is essential to remove unnecessary attributes and convert categorical values into numerical form.

```
data = data.drop(['id'], axis=1)
data['diagnosis'] =
data['diagnosis'].map({'M':1, 'B':0})
```

Inference: Removes irrelevant columns and converts labels into machine-readable numerical values.

C. Feature and Target Separation

Separating features and targets is necessary for supervised learning.

```
X = data.drop('diagnosis', axis=1)
```

Advantages: Easy to implement, works well for pattern recognition, and provides high prediction accuracy.

V. PERFORMANCE EVALUATION

The models are evaluated using accuracy, precision, recall, F1-score, confusion matrix, ROC curve, and AUC score.

A. Evaluation Metrics

The mathematical equations for each metric are defined as follows:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

VI. EXPERIMENTAL RESULTS

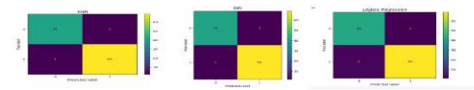
The experimental results show excellent performance for both Logistic Regression and KNN models. Table I presents a summary of prediction outcomes.

TABLE I
Comparison of Model Performance

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	98%–99%	High	High	High
KNN	98%–99%	High	High	High

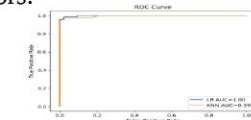
A. Confusion Matrix

The confusion matrix indicates that most predictions are correctly classified, with high diagonal values indicating strong classification performance and very few false predictions.



B. ROC Curve Analysis

ROC-AUC values close to 1.0 indicate excellent classification capability. AUC values between 0.98 and 1.00 demonstrate outstanding prediction quality for both models, which effectively distinguish malignant and benign tumors.



VII. ADVANTAGES OF THE PROPOSED SYSTEM

The proposed machine learning system provides several advantages:• High prediction accuracy

```
y = data['diagnosis']
```

D. Feature Scaling

Feature scaling standardizes the dataset and improves model performance.

```
from sklearn.preprocessing import
StandardScaler
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Advantages: Improves KNN distance calculations, helps Logistic Regression converge efficiently, and prevents feature dominance.

E. Feature Selection

Feature selection removes unnecessary attributes and improves prediction accuracy.

```
from sklearn.feature_selection import
SelectKBest, f_classif
selector = SelectKBest(score_func=f_classif,
k=5)
X_train = selector.fit_transform(X_train,
y_train)
X_test = selector.transform(X_test)
```

Inference: Reduces noise, improves model efficiency, and helps achieve high accuracy.

- Reduced human error
- Efficient feature selection
- Improved healthcare decision-making

VIII. LIMITATIONS

Despite high performance, the system has some limitations:

- Requires high-quality datasets
- Small datasets may reduce generalization capability
- KNN becomes computationally expensive for large datasets

IX. FUTURE ENHANCEMENTS

Future work can include:

- Applying deep learning techniques
- Using larger real-world medical datasets
- Developing real-time clinical applications
- Integrating cloud-based healthcare systems
- Comparing additional ML algorithms such as SVM and Random Forest

X. CONCLUSION

This paper presented a comparative analysis of Logistic Regression and K-Nearest Neighbors algorithms for breast cancer prediction. The proposed methodology included preprocessing, scaling, feature selection, and hyperparameter optimization. Experimental results demonstrated that both models achieved excellent performance with accuracy levels reaching approximately 98–99%.

The study confirms that machine learning algorithms can effectively support early breast cancer diagnosis and assist healthcare professionals in clinical decision-making. Proper preprocessing and feature engineering significantly improve prediction accuracy and overall system performance.

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